1. Python → App (Practical) — Git, IDE (VSCODE)
2. ML | DL | NLP | CV
3. GenAI

AI → Github Copilot } Industry

Coding assistant

1. Github copilot
2. Claude Code
3. OpenAI Codex

→ VS Code, Pycharm

{ CURSOR, Antigravity }

Basic coding task

{ VS Code + Github copilot } — Cleaner → copilot

{ vscode + Claude code }

Chapter

4.1, 4-0, GPT-5

Github copilot → (Free Acces) ← (Basic model)

Paid Acces (10$/month) ← Premuim model

Opus-6.5 GPT-5.3

INDUSTRY | PROJECT | LIVE CLASS

BASIC DOUBT ⇒ Installation, Code

1. Dashboard
2. G-sheet link | Github | Notion
3. Pre-requisite (Python)
   ML | DL | NLP | DS
4. VSCode + Copilot

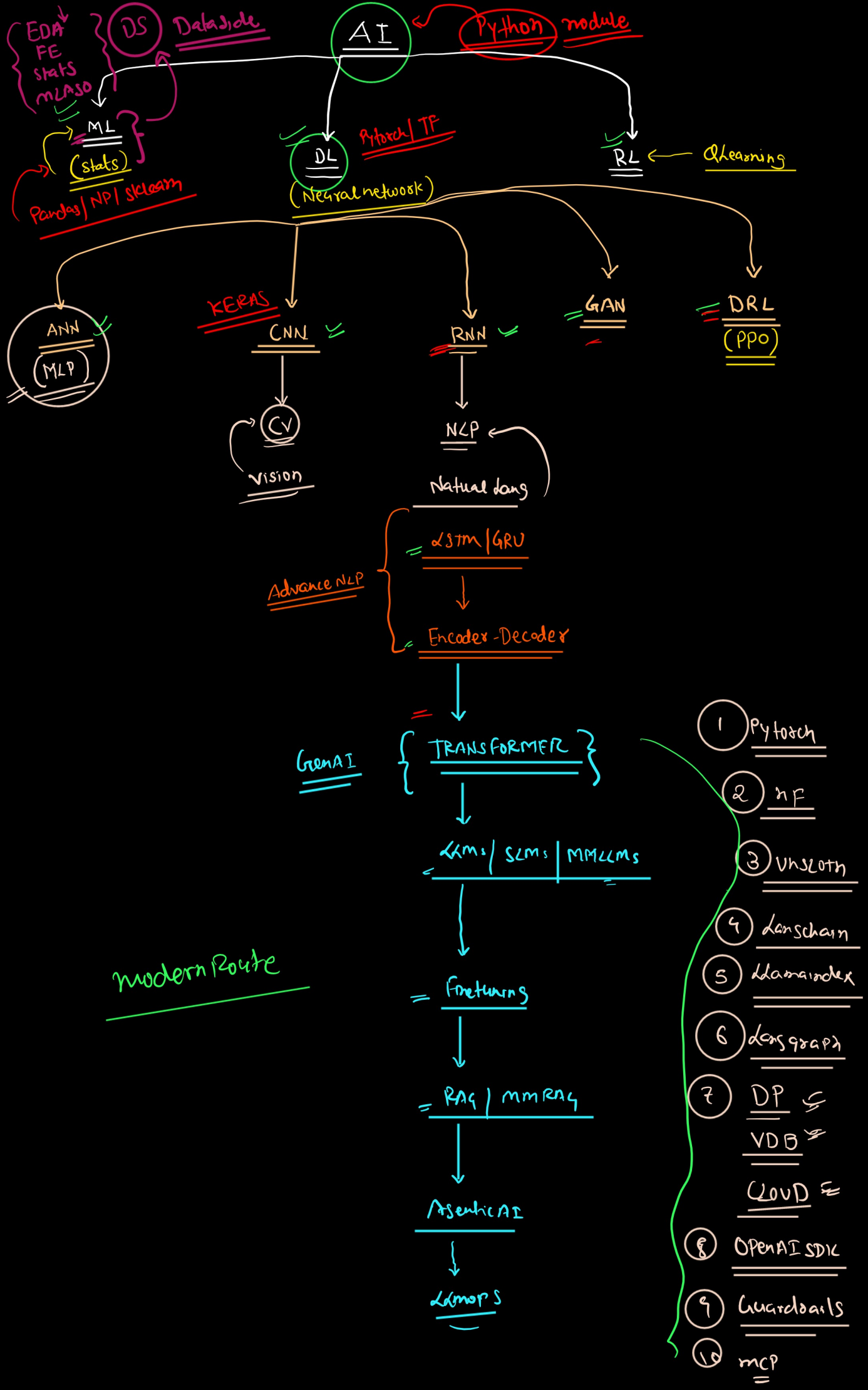
TODAY'S

1. GenAI
2. module -1
   encoding | embedding | tokenization
   Python

APP

Practical

DE


EDA
FE
stats
mLASO
DS
Dataside
AI
Python module
ML
(stats)
Pandas | NP | sklearn
DL
Pytorch | TF
(Neural network)
RL
Qlearning
ANN
(MLP)
KERAS
CNN
CV
vision
RNN
NLP
Natural lang
GAN
DRL
(PPO)
Advance NLP
LSTM | GRU
Encoder-Decoder
GenAI
TRANSFORMER
LLMs | SLMs | MMLLMs
modern Route
Fine tuning
RAG | MMRAG
Agentic AI
LLMOPS
1 Pytorch
2 HF
3 Unsloth
4 Langchain
5 Llamaindex
6 Langgraph
7 DP
VDB
CLOUD
8 OpenAI SDK
9 Guardrails
10 MCP

NLP → ~~RNN~~ → ~~LSTM | GRU~~

2018-2019

LLMs

ENCODING | Embedding ← TEXT

Vector | tokenization

NUMBERS

~~Encoder-Decoder Architecture~~ ← classical

TRANSFORMER -> Attention All you need

THEORY → BASE → INTERVIEW → LLM

encoding | embedding | transformer | LLM (SLM | mmLLM)

FT, RAG, Agentic AI, LLMOPS → POC | MVP

DP, VDB, GVD, MCP, cloud

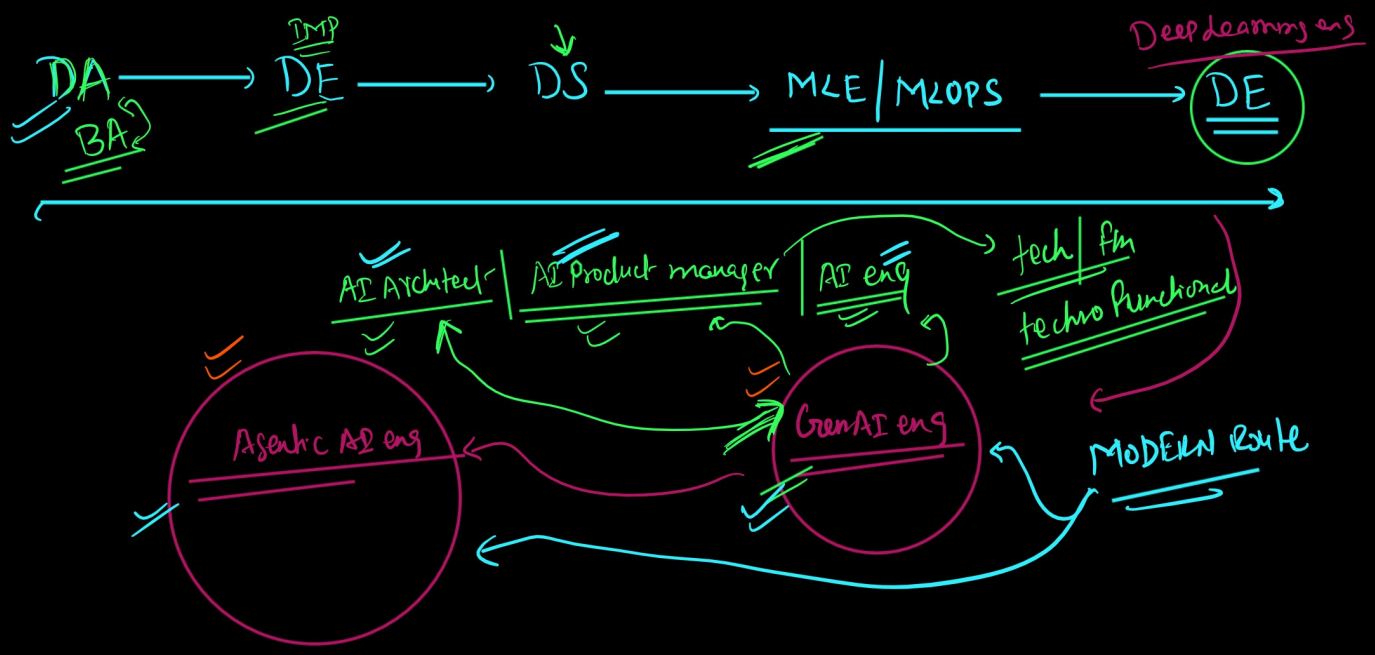
Proof of concept

AI-coding-Assistant ← Claude, copilot, OpenAI Codex

App → Dev, test, Deployment

POC → MVP → Full fledge dev

Scalable App



1. Encoding
2. Embedding
3. Semantic Search
4. KW search & semantic search
5. Application Encoding / embedding
6. models of embedding

vs vector

Practical

Any place in heart